Modeling the Meissner effect with differential equations

Research Question: How can differential equations be used to model and evaluate magnetic levitation in EDS Maglev trains?

Subject: Mathematics Modeling

Contents

1. Introduction

This Essay aims to model Maglev trains, focusing on their velocity, levitation height, and load size. Maglev trains are levitating trains which use little energy and got at incredibly high speeds, with the Japanese L0 series Maglev going at 603kmh (Brown, Dacquisto) (Japan Rail Pass).

Figure 1. A L0 Japanese Maglev train capable of going 603kmh (Singh)

My essay focuses on one specific type of maglev trains, EDS (electrodynamic suspension) maglev trains which use superconductive magnets to repel the magnetic track, allowing them to levitate (Fiveable).

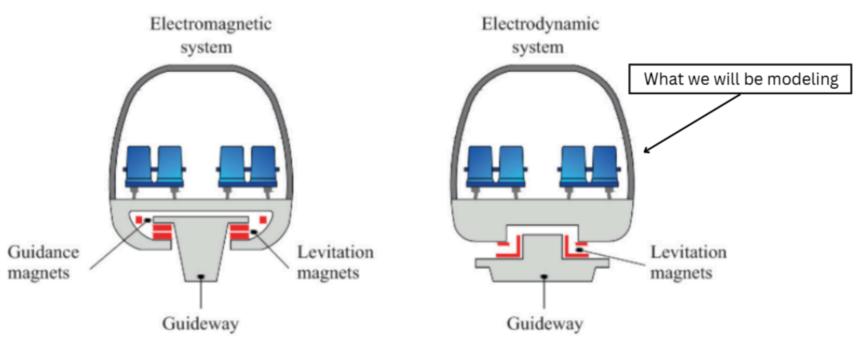

Figure 2. Diagram showing EDS maglevs in comparison to other maglevs (Guerrieri)

Once we model key variables such as Levitation height and speed, these values will be used to identify the real world applicability of these systems, modeling data such as the maximum number of passengers considered safe, and the voltage input rate for stable acceleration. As levitating trains are considered the future of transportation, understanding and modeling their performance is important to advance transportation innovation.

In my essay I start off by manipulating fundamental electromagnetic equations (such as the Fourier series and Lorentz force law) and Newton's laws using integration and differentiation to derive equations describing the behaviors of these trains. These equations will form ordinary differential equations (ODEs), equations involving a function and its derivative . After identifying key constants, I will apply Euler's method, a numerical technique that estimates solutions using small increments in time, to approximate these ODEs. I will then apply the Runge-Kutta methods, a more complex method of approximating solutions to ODEs, comparing results to understand the accuracy of these approximations. For this investigation my Research Question is, **"How can differential equations be used to model and evaluate magnetic levitation in EDS Maglev trains?"**

2. Mathematical key

2.1 Euler's Method

Euler's Method is a method of approximating ODEs. It specifically helps with differentials that give a relationship between a function and its derivative (like $f(x) = \frac{df}{dx}(x + 1)$). It does this by starting with an base x_0 value and a given $f(x_0)$ value, and then uses them to solve for the slope ($\frac{df}{dx_0}$) at this point (Dawkins). It then estimates the next point on the graph, of $f(x_1)$ (where $x_1 = x_0 + \Delta x$) by adding the $f(x)$ at the previous point, and the slope multiplied by the Δx, giving the equation:

$$f(x_1) = f(x_0) + \frac{df}{dx_0}\Delta x$$

By iterating through this process it can estimate the function based on a single starting value. This can be inaccurate (as shown in Figure 3) when done at higher Δx, due to errors being cumulative.

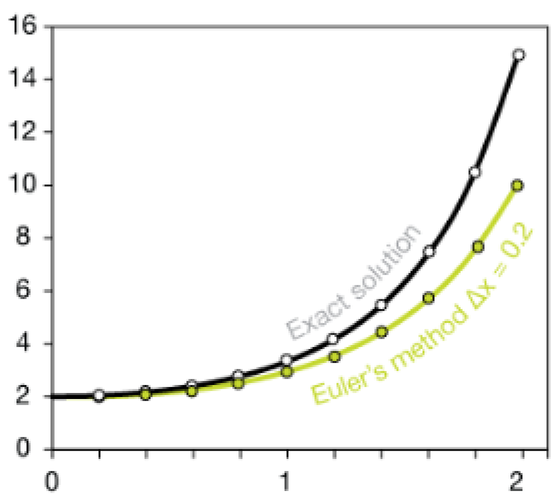

Figure 3. Example of implementation of Euler's method for an ODE (Cruzan)

2.2 Runge Kutta method

The Runge Kutta (RK) method is another way to approximate ordinary differential equations. It works by incorporating the slope over an interval rather than just the starting point (Neumann). To explain how this is done, we first have to set the first order ODE:

$$G(x, y) = \frac{dy}{dx} \text{ (this represents a first order ODE)}$$

Then we must set the step size, Δx, which is given by:

$$x_{n+1} = x_n + \Delta x$$

We then will apply Rk4's method of estimating y

$$y_{n+1} = y_n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4)$$

Where each of the k values is a predicted change in y given the slope of a point:

$$k_1 = \Delta x\, G(x_n, y_n)$$

$$k_2 = \Delta x\, G(x_n + \frac{\Delta x}{2}, y_n + \frac{k_1}{2})$$

$$k_3 = \Delta x\, G(x_n + \frac{\Delta x}{2}, y_n + \frac{k_2}{2})$$

$$k_4 = \Delta x\, G(x_n + \Delta x, y_n + k_3)$$

Here you can see that each $k = \Delta x \frac{dy}{dx}$, just the difference is that each tries to use the previous value to make differing estimations of what the slope is in between each interval, with k_2 and k_3 both trying to make predictions of what the midpoint derivative is.

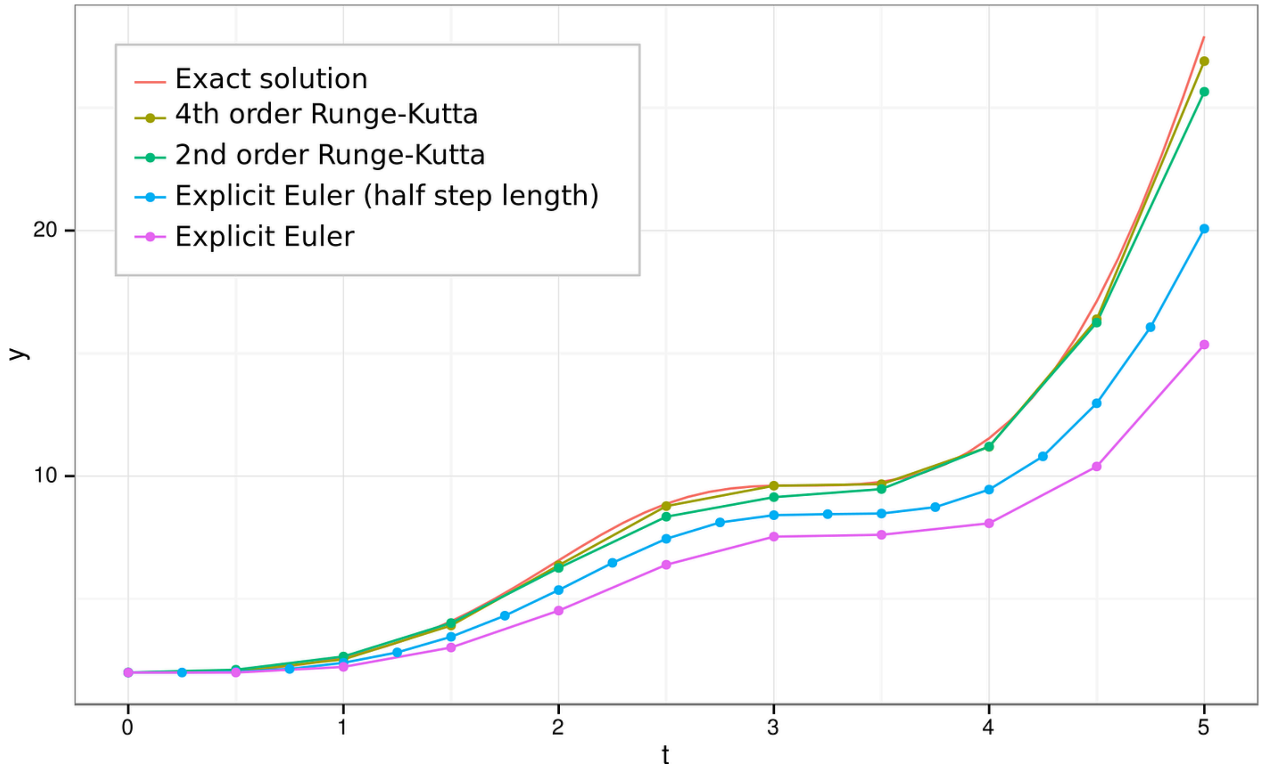

Figure 4. Comparison of Euler's method and RK4/2 methods (VisCircuit)

2.3 Del operator

The del operator (∇) is a vector differential in multivariable calculus, meaning it usually gives the gradient of a 3D function, represented as:

$$\nabla F(x,\ y,\ z) = \left(\frac{\partial F}{\partial x}, \ \frac{\partial F}{\partial y}, \ \frac{\partial F}{\partial z}\right)$$

Where "$\frac{\partial F}{\partial x}$" is a partial derivative, representing how the function F changes if keeping all values excluding x fixed. This is similar to a normal derivative, but where we assume all other variables (z and y) to be constants.

However in the case that $\nabla F(x,\ y,\ z)\ =\left(\frac{\partial F}{\partial x},\ 0,\ 0\right)$, then the gradient $F(x,\ y,\ z)$, is equivalent to the derivative, $F(x)$, represented by

$$\nabla F(x,\ y,\ z)\ = \frac{d}{dx}F(x)$$

It is also important not that the del operator can have different meanings, with $\nabla\ \times\ F$ giving the curl of the F function, which "measures the tendency of a field to rotate around a point" (Nykamp). $\nabla\ \cdot\ F$ gives divergence, meaning how much of the vector field is flowing outward from a point (Herman)

A key vector identity which I will use in calculating my model is the curl of a curl, given by:

$$\nabla\ \times\ \nabla\ \times\ F\ =\ \nabla(\nabla\ \cdot\ F)\ -\ \nabla^2 F$$

This will be important for my model.

2.4 Fourier Series

The fourier series is a series that sums up periodic waves to give us a set of nonuniform waves.

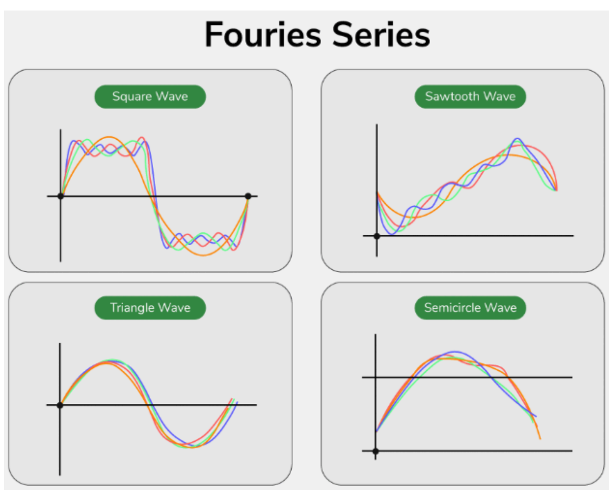

Figure 5. Different applications of the Fourier series (GeeksforGeeks)

It breaks down complex periodic functions into a sum of sines and cosines, making them easier to model. In my paper, this equation will be used to model the oscillating magnetic waves emitted by magnet arrays. This equation is:

$$f(x) = \frac{1}{2}A_0 + \sum_{n=1}^{N} A_n cos(k_n x) + B_n sin(k_n x)$$

A_0: The starting position of the series, 0 since we start from the beginning of the magnet.

A_n and B_n: Constants which get smaller for each n, in series that converge as $N \to \infty$.

k_n: is frequency, dependent on the period length and n.

As this series converges, we can approximate the $f(x)$ by stopping the summation at an earlier N value.

In my application of this equation we will use Parseval's theorem, which is:

$$\int_0^{\tau} \sum_{n=1}^{N} \left[A_n cos(k_n x) + B_n sin(k_n x)\right]^2 dx = \frac{\tau}{2} \sum_{n=1}^{N} \left(A_n^2 + B_n^2\right)$$

τ: A single period.

This can be shown by solving and using trig properties:

$$\int_0^{\tau} \sum_{n=1}^{N} \left[A_n cos(k_n x) + B_n sin(k_n x)\right]^2 = \int_0^{\tau} \sum_{n=1}^{N} \left[A_n^2 cos^2(k_n x) + 2A_n B_n sin(k_n x)cos(k_n x) + B_n^2 sin^2(k_n x)\right]$$

In this equation, trig properties give us that over a single period $sin(x)cos(x) = 0$, while $cos^2(x)$ and $sin^2(x)$ over a single period are both $= \frac{\tau}{2}$, thus:

$$\int_0^{\tau} \sum_{n=1}^{N} \left[A_n^2 cos(k_n x)^2 + 2A_n B_n sin(k_n x)cos(k_n x) + B_n^2 sin(k_n x)^2\right] = \sum_{n=1}^{N} \left(\frac{\tau}{2}A_n^2 + \frac{\tau}{2}B_n^2\right)$$

Here we can can take the $\frac{\tau}{2}$ out of the summation as it's a constant, giving us the final equation:

$$\int_{0}^{\tau} \sum_{n=1}^{N} \left[A_n cos(k_n x) + B_n sin(k_n x)\right]^2 = \frac{\tau}{2} \sum_{n=1}^{N} \left(A_n^2 + B_n^2\right)$$

3. Formulating the Mathematical Model

3.1 Electrodynamic suspension

EDS is a type of levitation based upon the meissner effect, where a superconductor forces the magnetic field inside it to be zero, creating a surface current that repels magnetic fields. This repulsion becomes stronger at high speeds, according to Faraday's law. The Lorentz law will be incorporated in the model to account for the strength of repulsion.The London equations will then be used to obtain the superconductor surface current.

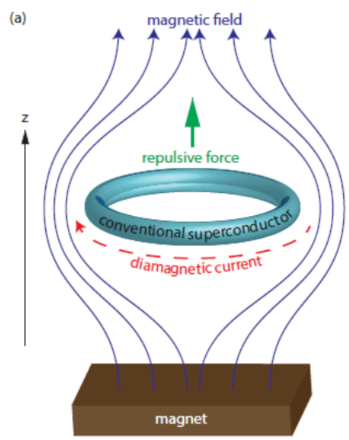

Figure 6. Perfect diamagnetism with z representing the levitation height (Kogar)

3.1.1 Repulsion with the Lorentz force law

The Lorentz force law is what gives the repulsion between superconductors and external magnetic fields:

$$F_{Repulsion} = \int^{V} J \times B(x,z)\, dV$$

J: Current density of eddy currents generated by the HTS

$B(x,z)$: The magnetic field from the magnet that reaches the surface of the HTS

V: The volume of the HTS

Integration over volume is the same as integration over all dimensions, making this the same as:

$$F_{Repulsion} = \int_0^l \int_0^w \int_0^{d_{sc}} J \times B_0(x, z)\, dz'\, dy\, dx$$

d_{sc} : The depth of the HTS over variable z' (depth into superconductor),

w: The width over y-axis,

l: length over the x-axis.

In EDS maglevs, the magnets are uniformly placed on the y-axis, causing the equation to be constant over this dimension. This allows for the simplification of the integral of with over y.

$$F_{Repulsion} = \left(\int_0^w dy\right)\left(\int_0^l \int_0^{d_{sc}} (J \times B(x, z)) dz'\, dx\right)$$

$$\int_0^w dy = w$$

$$F_{Repulsion} = w \int_0^l \int_0^{d_{sc}} J \times B\, dz'\, dx$$

As B only to the surface of the HTS, it is constant over z', leaving us with

$$F_{Repulsion} = w \int_0^l \left(\int_0^{d_{sc}} J\, dz'\right) \times B\, dx$$

Note: z and z' are two distinct values, z represents the distance between the superconductor and conductor, while z' represents depth into the superconductor.

3.1.2 Halbach arrays with Fournier series

Halbach array is an arrangement of permanent magnets that helps make the magnetic field much stronger on a single side.(K&J Magnetics).

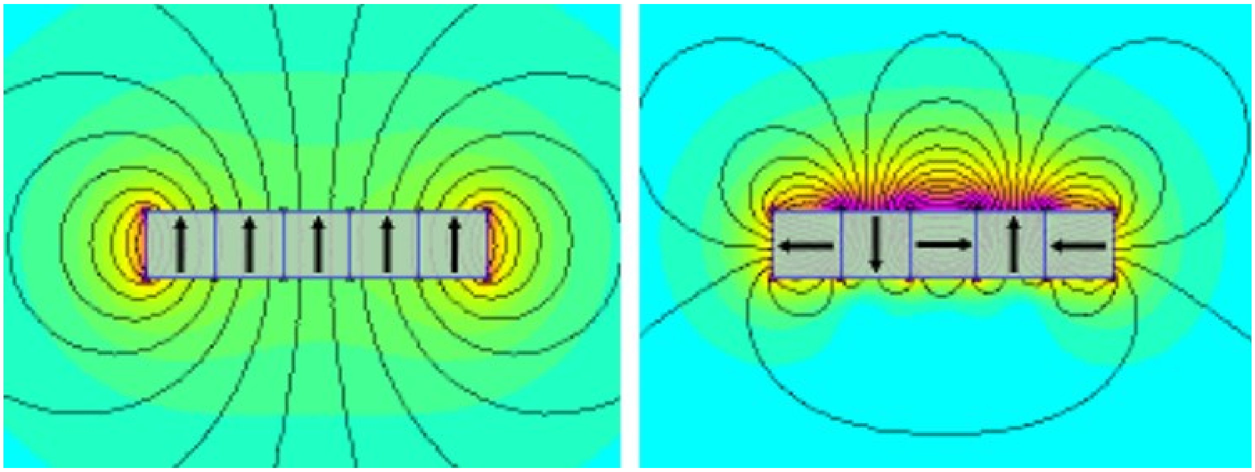

Figure 7. Halbach arrays compared to normal magnetic arrangements (Stanford Magnets)

These arrays create periodic magnetic waves. These magnetic waves can be approximated using the Fournier series (harmonics). This gives the magnetic field of the array as:

$$B(x) = \sum_{n=1}^{N}\left[A_n cos(k_n x) + B_n sin(k_n x)\right]$$

Note: This gives the field at the surface of the magnet, which will be represented as B_0 in the future

In magnets k_n is given by:

$$k_n = \frac{2\pi n}{\tau}$$

n: Term in the summation

τ: Period length (Length of repeating segment)

We also need to incorporate the z value, to show how the magnetic field gets weaker as distance increases. This is often done in magnets with Laplace transformation, where $e^{-k_n z}$ describes how the typical magnetic field decay works:

$$B(x,z) = \sum_{n=1}^{N} \left[A_n cos(k_n x) + B_n sin(k_n x)\right] e^{-k_n z}$$

Overall, this gives us how much of the magnetic field reaches the surface of the superconductor.

3.1.3 Applying the second London Equation for Field Decay

The second London equation describes superconducting density based on the magnetic field emitted by a superconductor, with a common form of the equation being:

$$\nabla \times J = -\frac{n_s e^2}{m} B$$

B: This is actually the general magnetic field in this case, different from the magnetic field reaching the surface of the superconductor

$\nabla \times$: The curl operator, explaining the currents curl around the superconductor.

n_s: the density of superconducting electrons.

e: The electron charge, a universal constant of 1.602×10^{-19} coulombs.

m: is the mass of a single superconducting electron, a constant depending on the material.

Now this equation can be manipulated by using Ampère's law:

$$\mu_0 J_s = \nabla \times B$$

$$J_s = \nabla \times \frac{B}{\mu_0}$$

μ_0: the constant for the permeability of free space

Note: Constants can be moved in and out of the curl operator ($\nabla \times$)

We can replace "J_s" in the London equation with Ampere's law, giving us.

$$\nabla \times \nabla \times \frac{B}{\mu_0} = -\frac{n_s e^2}{m} B$$

we can multiply the permeability of free space on both sides to move it over to the right portion of the equation.

$$\nabla \times \nabla \times B = -\frac{\mu_0 n_s e^2}{m} B$$

Additionally, we can use the vector identity of the curl of a curl below to replace the remaining left side of the equation:

$$\nabla \times \nabla \times B = \nabla(\nabla \cdot B) - \nabla^2 B$$

The divergence $(\nabla \cdot B)$ of a magnetic field is always zero, thus we are left with:

$$\nabla \times \nabla \times B = -\nabla^2 B$$

This gives us:

$$-\nabla^2 B = -\frac{\mu_0 n_s e^2}{m} B$$

$$\nabla^2 B = \frac{\mu_0 n_s e^2}{m} B$$

Here we can incorporate the London penetration depth equation.

$$\lambda_L = \sqrt{\frac{m}{\mu_0 n_s e^2}}$$

λ_L: Penetration depth of the superconductor (how far a magnetic field can penetrate the surface of the superconducting material till decaying to e^{-1} times the original field). We can substitute this into the equation with:

$$\frac{1}{\lambda_L^2} = \frac{\mu_0 n_s e^2}{m}$$

This gives us:

$$\nabla^2 B = \frac{1}{\lambda_L^2} B$$

From here we can specify that the value "B" is solely dependent on the z' (the depth into the superconductor). Additionally, as del operator (∇) of a singular vector is the same as a derivative, $\nabla^2 B$ becomes a double derivative $\frac{d^2 B(z')}{dz'^2}$ (meaning the rate of change of slope of magnetic field strength) giving us

$$\frac{d^2 B(z')}{dz'^2} = \frac{1}{\lambda_L^2} B(z')$$

This type of differential equation is approximated by assuming that B(z') has the exponential decay form of:

$$B(z') = C_1 e^{rz'}$$

C_1: In this scenario C_1 is the field strength at $z = 0$, which at the surface of the magnet (this is the same as B_0, and this will be substituted in replacement)

Using this equation we can solve for r by taking double derivative of B(z), which is $r^2 C_1 e^{rz'}$.

This gives us:

$$r^2 B_0 e^{rz'} = \frac{1}{\lambda_L^2} B(z')$$

And thus:

$$r^2 = \frac{1}{\lambda_L^2}$$

$$r = \frac{1}{\lambda_L} \text{ or } r = -\frac{1}{\lambda_L}$$

Because the strength of the field decays the further it gets from the surface, we will be using the negative exponent value, and by substituting all these values we get:

$$B(z') = B_0 e^{-\frac{z'}{\lambda_L}}$$

This is the field decay equation, giving us the strength of the magnetic field based on how deep into the superconductor we are.

$$B(z')_{internal} = B_0 e^{-\frac{z'}{\lambda_L}}$$

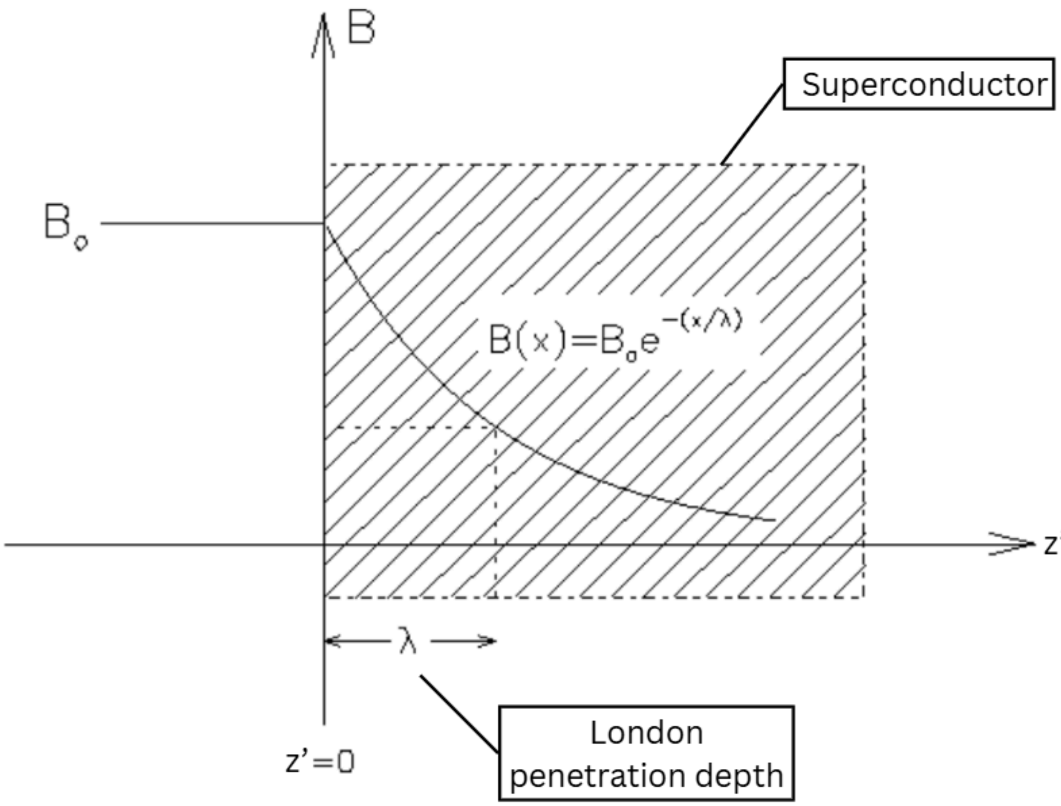

Figure 8. Penetration depth into a superconductor modeled (Sonier)

3.1.4 Applying the London Equations for Eddy Currents

To model the eddy currents, we must start with reuse a version of the second london equation:

$$J = -\frac{1}{\mu_0 \lambda_L^2} A$$

μ_0: penetration of the magnetic field in free space

λ_L:London penetration depth. A is the magnetic vector potential, given by:

$$B(z') = \nabla A$$

However, as explained earlier we are only accounting for the z' variable, giving us

$$B(z') = \frac{d}{dz'} A$$

Where we can take the integral of both sides to get

$$\int B(z') dz = A$$

Note: + C is not present, as C is assumed to be 0. This is because magnetic forces always must be able to decay to 0. This applies for the next integration as well.

Where as explained earlier, $B(z')_{internal}$ can be replaced with:

$$\int B_0 e^{-\frac{z'}{\lambda_L}} dz = A$$

Here we can take the integral, using chain rule in how it applies to $e^{z' \cdot a}$ (see previous **Note** for reason + C is not present)

$$-\lambda_L B_0 e^{-\frac{z'}{\lambda_L}} = A$$

We can plug this back into the original equation

$$J = -\frac{1}{\mu_0 \lambda_L^2} A$$

$$J = \frac{1}{\mu_0 \lambda^2} \lambda B_0 e^{-\frac{z'}{\lambda_L}}$$

$$J = \frac{B_0 e^{-\frac{z'}{\lambda_L}}}{\mu_0 \lambda}$$

This is the electrical current at any depth z'. To get the overall electric current we must take the integral of this electrical current over the superconductor's depth

$$J_{overall} = \int_0^{d_{sc}} J \, dz$$

$$J_{overall} = \int_0^{d} \frac{B_0 e^{-\frac{z'}{\lambda_L}}}{\mu_0 \lambda} dz'$$

$$J_{overall} = \frac{B_0}{\mu_0} - \frac{B_0 e^{-\frac{d}{\lambda_L}}}{\mu_0}$$

$$J_{overall} = - \frac{B_0}{\mu_0} (e^{-\frac{d}{\lambda_L}} - 1)$$

We can simply multiply the negative to the inside of the parentheses to get

$$J_{overall} = \frac{B_0}{\mu_0} (1 - e^{-\frac{d}{\lambda_L}})$$

This will be used in the final equation for the force of the levitation.

3.1.5 Incorporating Velocity in penetration depth

The penetration depth (λ) in the equations is also referred to as the "skin effect" for normal conductors, where a majority of the currents flow at the surface (and thus prevent the penetration of magnetic fields). This effect is given by the equation:

$$\lambda(t)_{SE} = \sqrt{\frac{\tau}{\mu \sigma \pi v(t)}}$$

Here, is magnetic permeability, and is electric conductivity (both constants dependent on material). is the length of a period (for the magnetic field attempting to penetrate). As the maglev's speed increases, this penetration depth switches over to the London penetration depth (λ_L). To model this shift we will be using the equation:

$$\lambda(t) = \sqrt{\lambda_L^2 + \left(\frac{\tau}{\mu\sigma\pi v(t)}\left(\frac{v_0}{v(t)}\right)^n\right)}$$

Here, v_0 is the speed at which the eddy currents start to become significant, and n is meant to model the rate of the gradual decay of the skin effect. This equation only applies when $v_0 \leq v$, otherwise we simply use the equation for the normal skin effect. This equation will be used to to model the penetration depth of the model

3.1.6 Combining forces

For this final equation, we must combine J and B into a singular equation

$$F_z = w\int_0^l \int_0^{d_{sc}} J \times B(x, z)\, dz'\, dx$$

However we already accounted for depth in , meaning the equation is now just:

$$F_z = w\int_0^l J_{overall} \times B(x, z)\, dx$$

As the two forces are perpendicular, their cross product is the same as just multiplying the values.

$$F_z = w\int_0^l J_{overall} \cdot B(x, z)\, dx$$

$$F_z = w\int_0^l \frac{B_0}{\mu_0}(1 - e^{-\frac{d}{\lambda_L}})B(x,z)\,dx$$

$$F_z = w\int_0^l \frac{B_0}{\mu_0}(1 - e^{-\frac{d}{\lambda_L}}) \cdot B_0 e^{-k_n z}dx$$

Where

$$B_0 = \sum_{n=1}^{N}\left[A_n cos(k_n x) + B_n sin(k_n x)\right]e^{-k_n z}$$

Thus substitution gives us:

$$F_z = w\int_0^l \frac{\sum_{n=1}^{N}\left[A_n cos(k_n x)+B_n sin(k_n x)\right]^2 e^{-k_n z}}{\mu_0}(1 - e^{-\frac{d}{\lambda_L}})dx$$

Since only the fourier series varies over the x variable, we can solely integrate this and take everything else out of the integration.

$$F_z = w\frac{(1-e^{-\frac{d}{\lambda_L}})}{\mu_0}\sum_{n=1}^{N} e^{-k_n z}\int_0^l\left[A_n cos(k_n x) + B_n sin(k_n x)\right]^2 dx$$

This simply repeats with every period, meaning we can simplify this to and integral of a single period, times Length divided by the length of a period (τ):

$$F_z = w\frac{e^{-2k_n z}(1-e^{-\frac{d}{\lambda_L}})}{\mu_0}\frac{l}{\tau}\sum_{n=1}^{N} e^{-2k_n z}\int_0^{\tau}\left[A_n cos(k_n x) + B_n sin(k_n x)\right]^2 dx$$

We can now apply Parseval’s theorem with the Fourier series, which gives us

$$\int_0^{\tau}\sum_{n=1}^{N}\left[A_n cos(k_n x) + B_n sin(k_n x)\right]^2 dx = \frac{\tau}{2}\sum_{n=1}^{N}\left(A_n^2 + B_n^2\right)$$

We can now substitute this into the original equation, giving us

$$F_z = w\frac{(1-e^{-\frac{d}{\lambda_L}})}{2\mu_0}l\sum_{n=1}^{N} e^{-2k_n z}\left(A_n^2 + B_n^2\right)$$

Now, to we must replace the London penetration depth with penetration depth being based on velocity, to account for the role of velocity in inducing stronger eddy currents:

$$\lambda(t) = \sqrt{\lambda_L^2 + \left(\frac{\tau}{\mu\sigma\pi}\left(\frac{v_0}{v}\right)^n\right)^2}$$

$$F_{Repulsion} = w\frac{(1-e^{-\frac{d}{\lambda(t)}})}{2\mu_0} l \sum_{n=1}^{N} e^{-2k_n z}\left(A_n^2 + B_n^2\right)$$

This is the final equation, which gives the total force from the repealed magnetic waves (which reach the superconductor. Additionally, for further simplification we can multiply length and width for applicable Area (A_{area}) which the superconductor works over

$$F_{Repulsion} = \frac{(1-e^{-\frac{d}{\lambda(t)}})}{2\mu_0} A_{area} \sum_{n=1}^{N} e^{-2k_n z}\left(A_n^2 + B_n^2\right)$$

This is the equation that we will use for the model to give the lift force.

`3.1.7 Deriving for Acceleration

Now we can create the second order ordinary differential equation by using Newton's second law, which describes acceleration depending on an applied force. This equation is

$$ma = F_{net}$$

m: mass of the train

a: acceleration.

Acceleration in this scenario is in the z direction (levitation height), thus it is:

$$a = \frac{dz^2}{d^2t}.$$

In the scenario of levitation, there are two values that make up the total Force. These are

1. The previously calculated repulsion and 2. Gravity. The force of gravity is represented by mg, where g is the gravitational acceleration (dependent on location) which is $-\ 9.807\frac{m}{s^2}$. Plugging in these two values, replacing acceleration, and dividing by mass on both sides gives us

$$\frac{dz^2}{d^2t} = \frac{F_{Repulsion}}{m} - 9.807\frac{m}{s^2}$$

If $\frac{F_{diamagnetic}}{m} \leq 9.807\frac{m}{s^2}$ in the entire period modeled, then $z = 0$, then the acceleration is just $\frac{dz^2}{d^2t} = 0$. This is almost the final equation for the diamagnetic levitation. We lastly need to incorporate $v(t)$ and $x(t)$ to finish the model.

3.2 Linear Synchronous Motors

LMS (Linear Synchronous Motors) motors are a type of motor that generates linear motion, rather than rotational motion like in normal motors (Zoke Automation). To explain how they work, we first have to examine normal motors. Normal motors run electricity through a coil between two permanent magnets to repel and attract simultaneously, switching the current's direction continuously so that it keeps moving in this direction.

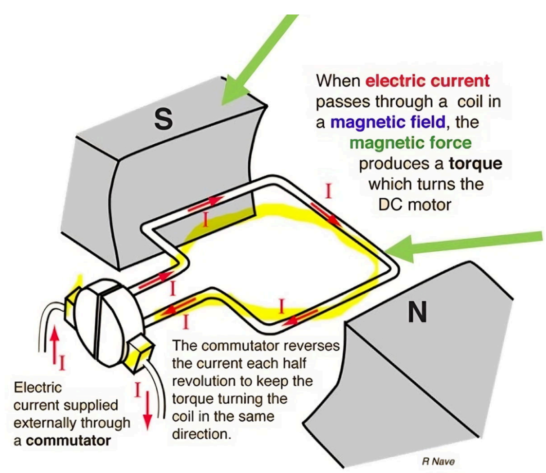

Figure 9. How a normal motor works by flipping the direction of the electrical current (Magnetic Innovations)

. A Linear motor works with the same principle, just applying it to magnets forward.

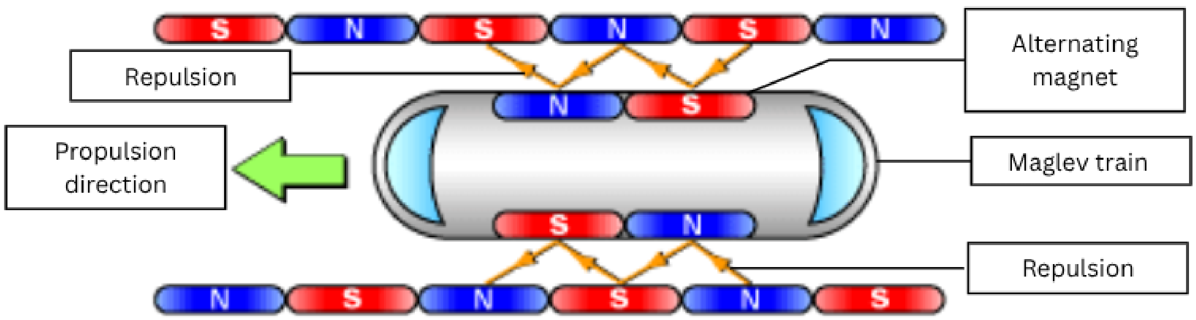

Figure 10. Linear Motor shown with a maglev train (Hamed)

For this model we will be calculating the force of this motor using the Lorentz force equation and accounting for the drag to derive overall forward motion.

3.2.1 Motor Force calculations

This motor applies Newton's second law to identify the motion caused by related forces, giving

$$m\frac{dv}{dt} = F_{motor}(t) - F_{drag}(t)$$

To find $F_{motor}(t)$ We start with the Lorentz law for the motor winding equation. This equation gives

$$F_{motor}(t) = k_t I(t)$$

k_t: The Thrust constant, converting current into thrust force.

$I(t)$: Current

To derive the current we must use formula for time domain Kirchhoff-voltage law, giving the voltage's relation to current

$$V(t) = L_m \frac{dI}{dt} + R_m I(t) + \frac{d\psi}{dt}$$

L_m: The phase inductance (constant)

R_m: magnetic resistance of the motor.

ψ: magnetic flux,

Magnetic flux has a linear relationship with displacement, thus we can split the derivative into $\frac{d\psi}{dx}\frac{dx}{dt}$ with chain rule, and as the equation is linear, $\frac{d\psi}{dx}$ is a constant, commonly represented as k_e. Combining this with the fact that the derivative of displacement over time is velocity,

$$\frac{d\psi}{dt} = k_e v.$$

$$V = L_m \frac{dI}{dt} + R_m I(t) + k_e v$$

To solve for $I(t)$ we can move current related portions to one side, and dividing both sides by L_m

$$\frac{dI}{dt} + \frac{R_m}{L_m} I(t) = \frac{1}{L_m}\left(V - k_e v\right)$$

We must now multiply both sides by $e^{\frac{R_m}{L_m}t}$, because this allows us to simplify our left side into a single derivative (a method called integration using a factor)

Giving us…

$$\frac{dI}{dt}e^{\frac{R_m}{L_m}t} + \frac{R_m}{L_m}I(t)e^{\frac{R_m}{L_m}t} = \frac{d}{dt}\left(I(t)e^{\frac{R_m}{L_m}t}\right) = e^{\frac{R_m}{L_m}}\frac{1}{L_m}\left(V - k_e v\right)$$

$$\frac{d}{dt}\left(I(t)e^{\frac{R_m}{L_m}t}\right) = e^{\frac{R_m}{L_m}t}\frac{1}{L_m}\left(V - k_e v\right)$$

Here we will integrate both sides, and by also taking the constants outside of the integral, we get

$$I(t)e^{\frac{R_m}{L_m}t} = \frac{1}{L_m}\int_0^t e^{\frac{R_m}{L_m}t}\left(V - k_e v\right)ds$$

$$I(t)e^{\frac{R_m}{L_m}t} = \frac{1}{R_m}\left(V - k_e v\right)(e^{\frac{R_m}{L_m}t} - 1)$$

Finally, we can divide the e term on both sides to give us:

$$I(t) = \frac{1}{R_m}\left(V - k_e v\right)(1 - e^{-\frac{R_m}{L_m}t})$$

From here we can just substitute this into the motor equation, to get

$$F_{motor}(t) = k_t\frac{1}{R_m}\left(V - k_e v\right)(1 - e^{-\frac{R_m}{L_m}t})$$

Note: Voltage and velocity are treated as constant because they change much more slowly than the current, so their variation is negligible while the current adjusts.

3.2.2 Drag calculations

To calculate F_{drag} We must account for two types of drag, aerodynamic drag, which gives the drag due to movement through air, and the eddy drag (also known as magnetic damping), a force that opposes motion in a magnetic field.

$$F_{drag} = F_{aero} + F_{eddy}$$

The aerodynamic drag (F_{aero}) is given by the equation…

$$F_{aero} = \frac{1}{2}\rho C_d A_f v(t)^2$$

ρ: A constant for the density of the fluid which the object passes through (in this case it is air)

C_d: drag coefficient, dependent on the shape and surface of the train.

A_f: frontal area of the train facing the flow of air

For the eddy force, t is proportional to velocity at lower speeds (giving us $C_1 v(t)$). Additional drag related to the skin effect, will be removed for simplification. Here C_1 is a constant to be calculated. By incorporating this and the F_{aero}, we get the overall drag force:.

$$F_{drag} = C_1 v(t) + \frac{1}{2}\rho C_d A_f v(t)^2$$

By adding both derived F_{drag} and F_{motor} values into this original acceleration function gives us

$$m\frac{dv}{dt} = k_t \frac{R_m}{L_m^2}\left(V - k_e v\right)\left(1 - e^{-\frac{R_m}{L_m}t}\right) - C_1 v(t) - \frac{1}{2}\rho C_d A_f v(t)^2$$

This gives us a first order ordinary differential equation, requiring us to assign a function to $V(t)$ to be properly solved for. The most likely and simplest way model $V(t)$ in real trains an exponential function, as these trains smoothly increase voltage, like:

$$V(t) = V_{max}(1 - e^{-\frac{t}{r}})$$

Where V_{max} is the maximum reached voltage, and r is a chosen constant to model the rate at which this exponential increase occurs. After dividing mass on both sides, this gives us

$$\frac{dv}{dt} = \frac{1}{m}\left(k_t \frac{R_m}{L_m^2}\left(V_{max}(1 - e^{-\frac{t}{r}}) - k_e v\right)\left(1 - e^{-\frac{L_m}{R_m}t}\right) - C_1 v(t) - \frac{1}{2}\rho C_d A_f v(t)^2\right)$$

This final equation will be used to find velocity over time, depending on the max voltage chosen.

4. Creating the Model

The model we will create will have specifications and materials based on the Japanese L0 series maglev, specifically the one designed for the Chuo Shinkansen line. These use NbTi (niobium titanium) superconductors, and are currently the fastest Maglev trains (Stanford). When creating graphs for our function, we will use the VS code platform.

4.1 Identifying all Constants

4.1.1 Velocity Constants

For the velocity model ,we have 6 constants which are defined in the appendix. To find k_t and k_e we must plug these values in, and consider the fact that that in optimal scenarios

$$k_t \approx k_e \approx k$$

This is due to conservation of energy in an ideal motor, where electrical power equals mechanical power. Using the maximum velocity of $167.5\frac{m}{s}$ (converted from 602kph) and plugging in all constants we get the equation:

$$0 = \left(k\frac{1}{R_m}(V - kv)(1 - e^{-\frac{R_m}{L_m}t}) + C_1 v(t) + \frac{1}{2}\rho C_d A_f v(t)^2 + C_3 v(t)^3\right)$$

$$\lim_{t\to\infty} 1 - e^{-\frac{R_m}{L_m}t} = 1$$

$$0 = \left(k\left(\frac{20{,}000 - k167.5}{0.1}\right) ds - 29(167.5) - 1.3628125(167.5)^2\right)$$

$$0 = -k^2 335 + k40000 - 43092.9082031$$

Using the quadratic formula, we get

$$k = 21.32674$$

$$k \approx k_t \approx k_e \approx 21.32674$$

Now we have all necessary constants.

4.1.2 Levitation Constants

For levitation, we need A_{area} which the superconductor acts on, and n in the penetration value which gives the rate at which the skin depth penetration decreases as velocity increases. We will use the point of maximum levitation to calculate constants, where:

$$0 = \frac{F_{Repulsion}}{m} - 9.807$$

Where $F_{Repulsion}$ is:

$$F_{Repulsion} = \frac{(1-e^{-\frac{0.01}{\lambda_L}})}{2\mu_0} A_{area} \sum_{n=1}^{N} e^{-2k_n \times 0.1}\left(A_n^2 + B_n^2\right)$$

$$F_z = \frac{(1-e^{-\frac{0.01}{0.0000001}})}{2\mu_0} A_{area} \sum_{n=1}^{N} e^{-2k_n \times 0.1}\left(A_n^2 + B_n^2\right)$$

Plugging in the fourier series for N=5, and approximating the top portion of my fraction, we get the equation:

$$F_z = \frac{(1-e^{-\frac{0.01}{0.0000001}})}{2\mu_0} A_{area} 0.117$$

$$F_z = 3.98 \times 10^5 A_{area} 0.117$$

$$F_z = 4.6566 \times 10^4 A_{area}$$

Plugging this back into the previous equation gives us:

$$9.807m = 4.6566 \times 10^4 A_{area}$$

Plugging in the value for mass, we can solve for A_{area} as

$$9.807 \times 362874 = 4.6566 \times 10^4 A_{area}$$

$$3558705.318 = 4.6566 \times 10^4 A_{area}$$

$$76.4444613372 = A_{area}$$

This means the effective area we are working over is $\approx 76.444\, m^2$

Now to solve for the penetration decay rate n we will apply the equation with the speed at which levitation starts ($30ms^{-1}$) and first solve for penetration depth. As levitation starts occurring at these speeds, we can assume that the force is equal to mass times gravity at $z = 0$:

$$9.807m = F_{Repulsion} = \frac{(1-e^{-\frac{0.01}{\lambda(t)}})}{2\mu_0} A_{area} \sum_{n=1}^{N} e^{-2k_n \times 0}\left(A_n^2 + B_n^2\right)$$

We can now calculate the sum of the Fourier series, and simplify, giving:

$$9.807m = \frac{(1-e^{-\frac{0.01}{\lambda(t)}})}{2\mu_0} 76.423 \cdot 1.54303$$

$$9.807m = (1 - e^{-\frac{0.01}{\lambda(t)}}) \cdot 397887.358184 \cdot 76.444 \cdot 1.54303$$

$$9.807m = (1 - e^{-\frac{0.01}{\lambda_L}}) \cdot 46932956.6486$$

Substituting for math, we get:

$$9.807 \times 362874 = (1 - e^{-\frac{0.01}{\lambda(t)}}) \cdot 46932956.6486$$

$$3558705.318 = (1 - e^{-\frac{0.01}{\lambda(t)}}) \cdot 46932956.6486$$

$$0.92417470426 = e^{-\frac{0.01}{\lambda(t)}}$$

$$ln(0.92417470426) =- 0.07885415133 =- \frac{0.01}{\lambda_L}$$

$$0.12681640511 = \lambda(t)$$

Now we can substitute this into the previous equation for penetration depth

$$0.12681640511 = \sqrt{\lambda_L^2 + \left(\frac{\tau}{\mu\sigma\pi}\left(\frac{v_0}{30}\right)^n\right)^2}$$

$$0.0160824006 = \lambda_L^2 + \left(\frac{\tau}{\mu\sigma\pi}\left(\frac{v_0}{30}\right)^n\right)^2$$

Here, λ_L^2 is so small, it is essentially ≈ 0 (approximately 10^{-14}), so we can ignore it:

$$0.12681640511 = \left(\frac{\tau}{\mu\sigma\pi}\left(\frac{v_0}{30}\right)^n\right)$$

Now, substituting all other values and dividing, we are left with:

$$0.12681640511 = 0.12665147955\left(\frac{v_0}{30}\right)^n$$

$$1.00130220003 = \left(\frac{v_0}{30}\right)^n$$

From here we can see 30m/s is a good value for v_0. However, we also know that at the max velocity (168m/s) we should have:

$$10^{-9} \geq \left(\frac{\tau}{\mu\sigma\pi} \left(\frac{v_0}{v} \right)^n \right)$$

Because this is where the skin depth becomes less than a single nm (london penetration depth is 100nm), becoming insignificant. So using 30m/s as a v_0, we can now find the minimum n for the model to be proper.

$$10^{-9} \geq 0.12665147955\left(\frac{30}{168}\right)^n$$

$$7.89568352 \cdot 10^{-9} \geq \left(\frac{5}{28}\right)^n$$

$$\log_{\frac{5}{28}}(7.89568352 \cdot 10^{-9}) \leq n$$

Note: Here the inequality sign flipped because we took a log where the base is < 1.

$$\log_{\frac{5}{28}}(7.89568352 \cdot 10^{-9}) \leq n$$

$$10.82964438 \leq n$$

This means $11 = n$ would be acceptable for modeling the performance of the model

Now we have all of the constant values, meaning we can make the model.

4.2 Modeling Velocity

We will start out by observing the velocity of the maglev over time, comparing RK4 and Euler's method with a step size of 0.1 (it calculates velocity at each $\Delta t = 0.1$) over the first 300 seconds

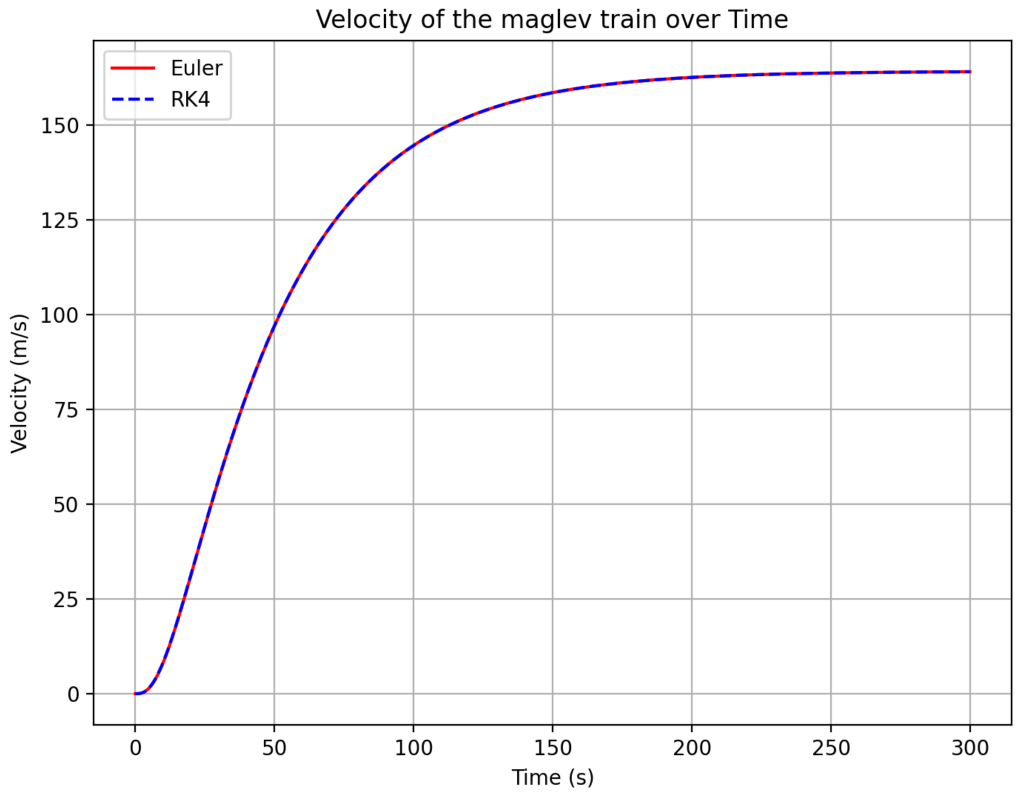

Figure 11. Velocity of Maglev within first 300 seconds with step size 0.1 second

The model is accurate, as the speed plateau at 167.5 m/s (603kph) which is the max speed of an L0 maglev, and it takes 4 minutes (240 seconds) to reach the maximum, which is how long it takes an L0 to accelerate to max speed. Here it can also be seen that the RK4 method and Euler method are the same. Now we will look at the relation between voltage and velocity.

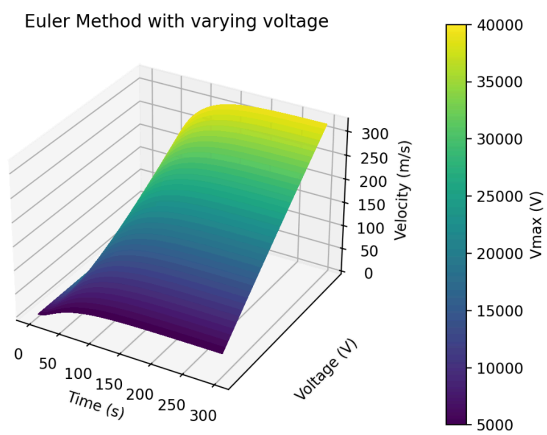

Figure 12. Velocity based off input Voltage (VSCode)

As seen in figure 13, the higher the voltage, the higher the final velocity reached, however, this model doesn't account for the Maglev's stability, as at high velocities the maglev could tip over while taking turns. Finally, we will also try to understand the impact higher mass has on the Maglev's acceleration.

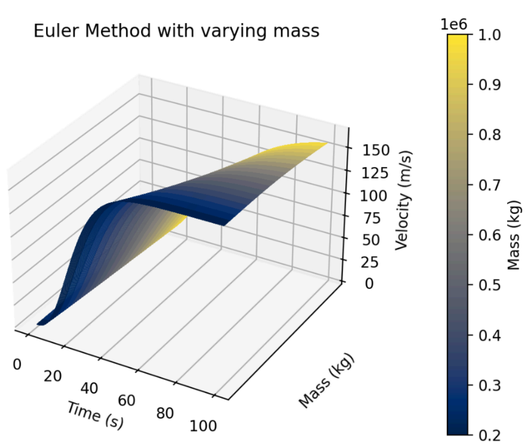

Figure 13. Velocity of Maglev depending on its mass

Here we can see that the mass of the maglev mainly only affects the time it takes to reach a certain velocity, but doesn't actually affect the end behavior of the graph (as $t \Rightarrow \infty$). Lastly we need how the rate at which the voltage is ramped up can affect the acceleration.

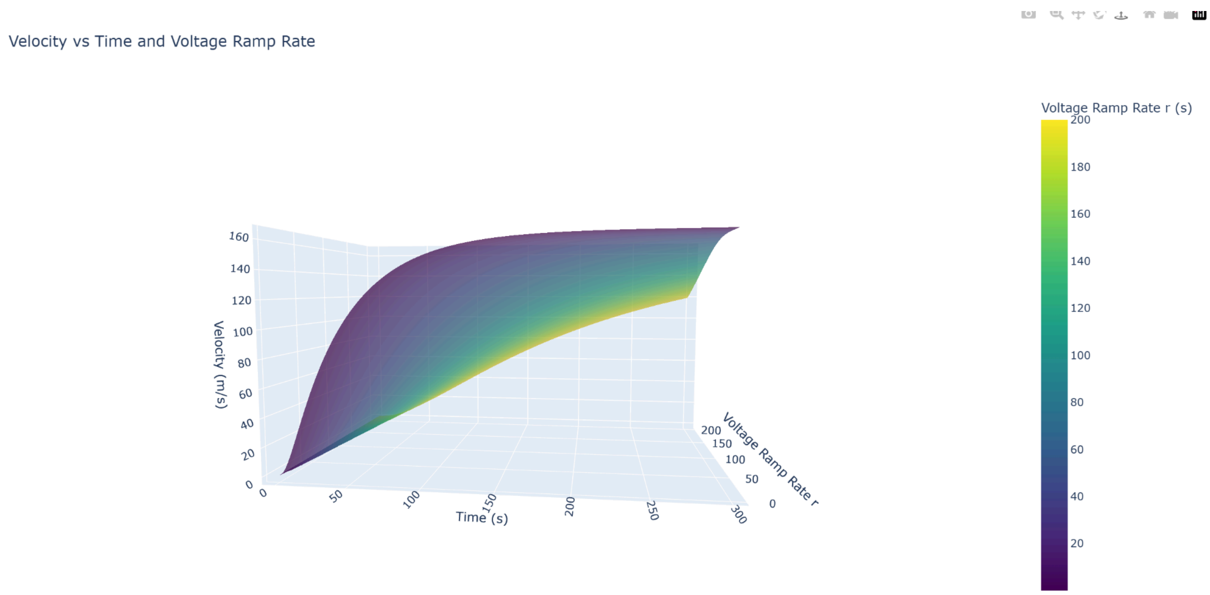

Figure 14. Velocity based off voltage ramp rate r

Here, there are different voltage ramping rates affect the rate at which and time taken to reach the maximum velocity, where the higher the r value, the lower maximum acceleration. This is shown in figure 16

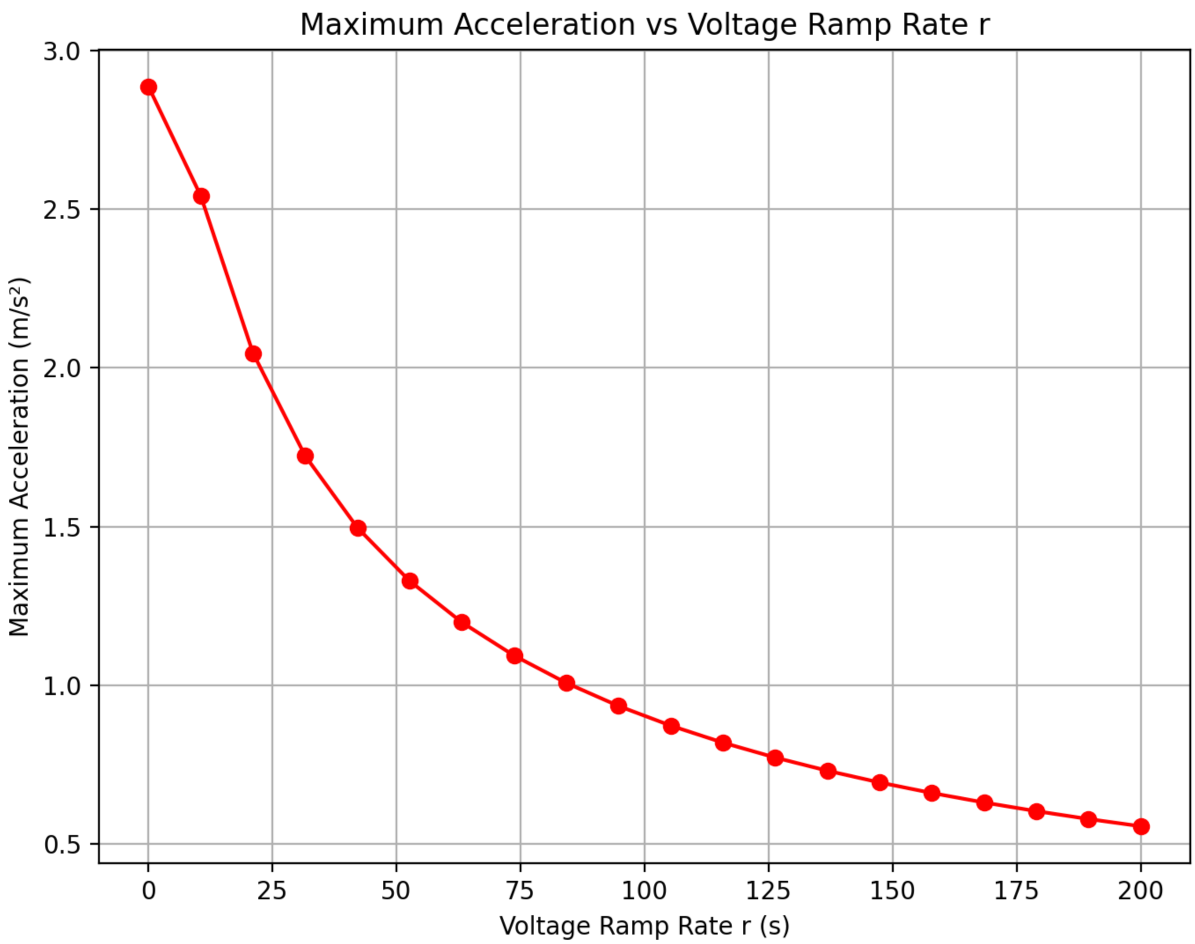

Figure 15. Max acceleration based off voltage ramp rate

The higher the Ramp rate, the lower the maximum acceleration. This is relevant, as a high acceleration can be dangerous to passengers.

4.3 Modeling Levitation

To model levitation, we will first use the equation to graph the height of the L0 over time, using the previously calculated velocity values (using $\Delta t = 0.1$).

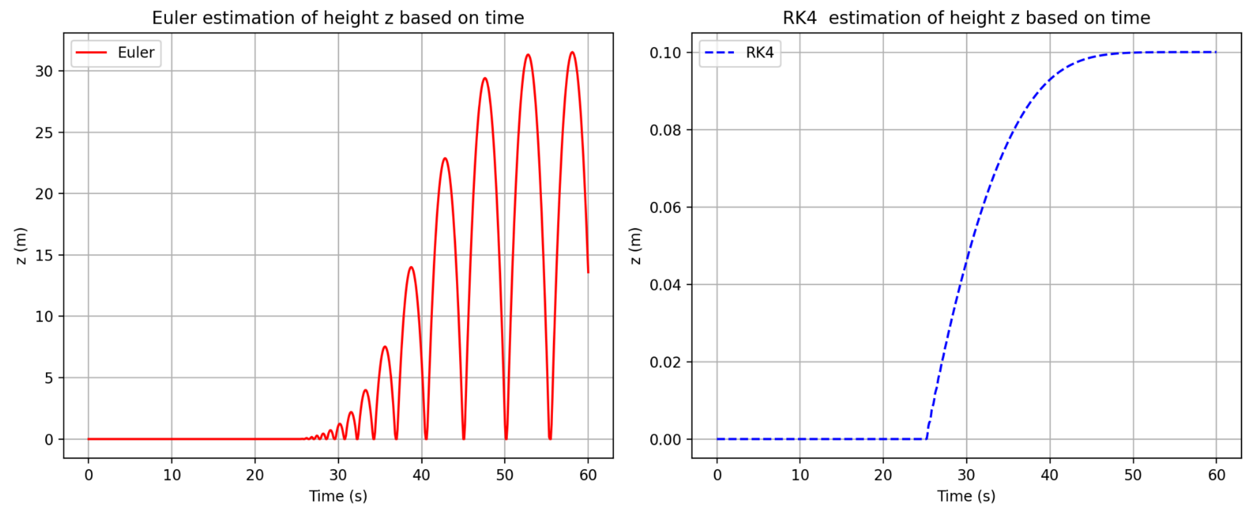

Figure 16. Levitation in first minute with step size of 0.1

Here, Euler's method shows oscillation within the levitation, while RK4 smoothly predicts levitation. This is because there is extremely slight oscillation caused by the counteracting forces, and while Euler's method uses the acceleration at a single point (thus overexpressing this oscillation), RK4 is able to sum up and average out the oscillations, canceling them out, making predictions smooth. This is also attributed to "Global error", where the error of RK4 is proportional to the step size to the fourth power, being smaller due to splicing the step size 4 more times.

At higher step sizes, we can see that the oscillations become more accurate, as seen in figure 18.

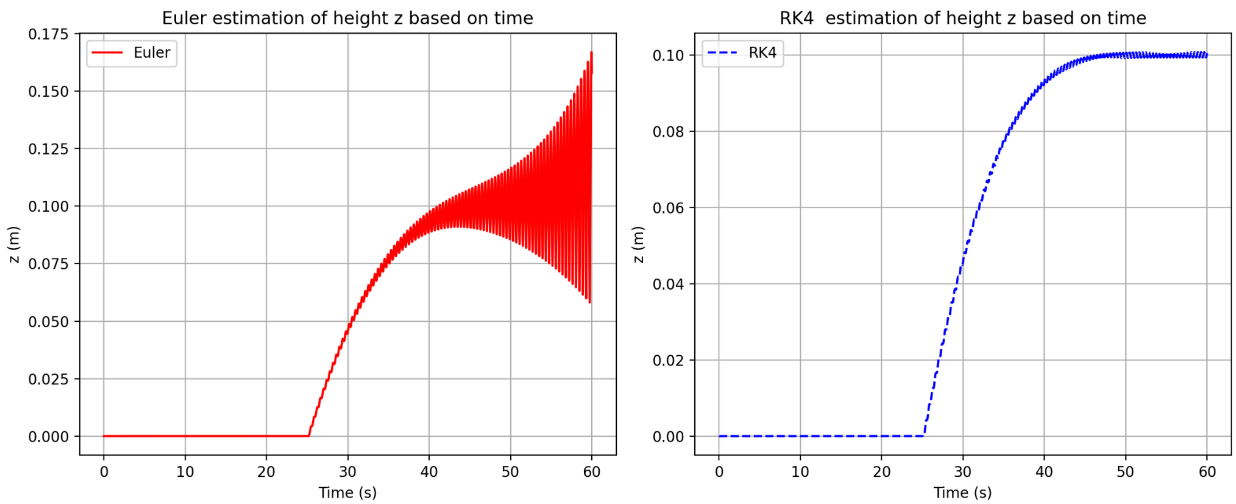

Figure 18. Levitation height in first minute based on step size of 0.001 (VSCode)

Here we can see that with larger step size, the oscillation in the model becomes more frequent, but also reduces the amplitude of these oscillations, likely because of how the smaller step size can constantly account for how the oscillation changes. Overall RK4 is more accurate at representing this levitation height, as only very small oscillation actually occurs in the real world.

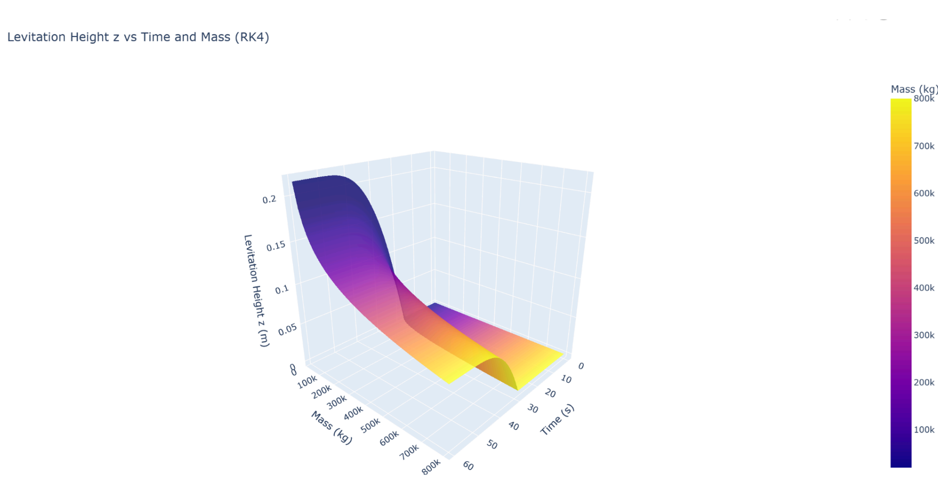

Figure 17. Levitation height in first minute based off mass

At higher masses, the final levitation height is lower. At ~460,000kg (460 metric tons) the levitation height is 9cm, and at ~600,000kg (600 metric tons) the levitation height is 8cm. It also takes slightly longer for levitation to start as the mass increases.

4.4 Discussion and Application

The model gives 2 key safety parameters for the Maglev train. First, the rate at which the voltage can ramp without high acceleration that causes passengers to experience jerk/injury. The safety standard for the maximum acceleration is usually $1.0\frac{m}{s^2}$, (acceleration of $3.6kph$) (Powell). The r which has a maximum acceleration of $1.0\frac{m}{s^2}$ is ≈ 85, which means at $r \geq 85$ the voltage ramping is safe for the passenger.

Secondly, the mass carried is an important parameter for safety, as higher mass can lower the levitation to dangerous heights. The danger levels we will be considering are moderate (9cm levitation) and high (8cm levitation) . The overall kg limit at moderate is 460,000kg. As the train itself is 360,000kg, leaving 100,000kg for storage and passengers. As the average person weighs 62kg, we can create the equation below. (Walpole).

$$100,000kg = N_{of\ passengers} \cdot 62kg + Mass_{of\ storage}$$

If we are taking the more dangerous (high level) levitation height, then we can store an additional 140,000kg on this train.

5. Conclusion

In our study we found that differential equations were highly successful at modeling the meissner effect (levitation) and the overall Maglev performance. This was because the analysis not only gave us data similar to real world scenarios, but we also were able to derive key parameters that kept the Maglev within a safe levitation range, and prevented passenger danger or discomfort by reducing the maximum acceleration. We also identified that RK4 was far more accurate at modeling levitation height because of how Euler's method would over express oscillation.

This approach also had a couple limitations that weakened the model. We do not account for several phenomena and real world scenarios, we also oversimplify geometry (as many arrangements aren't flat surfaces) and don't account for temperature/heat. To improve the study, we could also incorporate the temperature values though London penetration depth, as this constant increases as temperature increases. We can also extend the study to understanding the energy and financial costs of these systems, modeling this integrating voltage and incorporating the costs of cryogenic (cooling) systems.

The study identified how differential equations could help enumerate the magnetic forces of Maglev trains, thus allowing us to model their levitation and movement. This answers the research question, as differential equations can be used to model and evaluate magnetic levitation in EDS maglev trains to a great extent.

6. Appendix

Table 1. Coefficients for the Velocity equation

Symbol	Name	Value	Units
m	Mass	326587(12 cars) -435449 (16 cars)	kg
L_m	Motor Inductance	3.3	H
ρ	Air Density	1.225	$\frac{kg}{m^3}$
A_f	Frontal Area of Train	8.9	m^2
C_d		0.25	Non-dimensional
C_1	Eddy drag Coefficent	29	N
k_t	Thrust Coefficient	236.62284	$\frac{N}{A}$
k_e	Faraday effect Coefficient	236.62284	$\frac{V}{ms^{-1}}$
V_{max}	Maximum voltage	Varies	V

Table 2. Coefficients for the Levitation equation

Symbol	Name	Value	Units	Source
m	Mass	326587(12 cars) -435449 (16 cars)	kg	
A_{area}	Working area of the	$-$	m^2	Based on design

	superconductor			
A_n	Fourier coefficient for Halbach array	Is defined for each harmonic in Table 3	T	
B_n	Fourier coefficient for Halbach array	Is defined for each harmonic in Table 3	T	
d	Depth of superconductor	0.01	m	
k_n	The wavenumber of each harmonic	Is defined for each harmonic in Table 3	m^{-1}	
τ	Period length of halbach array	0.5	m	
μ_0	Permeability of free space	$4\pi \times 10^{-7}$	Hm^{-1}	
μ	Magnetic permeability of material	$\approx \mu_0$	Hm^{-1}	
σ	Conductivity of material	10^5	S/m	
λ_L	London Penetration Depth	0.0000001	m	
A_{area}	Effective area which the repulsion works on	76.444	m^2	
v_0	Velocity which eddy currents become effective	30	ms^{-1}	
n	Decay of skin effect at higher velocity	11	Non-dimensional	

Table 3. Coefficients for the fourier equation

Harmonic Number	k_n	A_n	B_n
1	12.566	0	1.2
2	25.133	0.01	0.3
3	37.699	0.005	0.1
4	50.265	0.002	0.05
5	62.832	0.001	0.02

7. Bibliography

Brown, Terry, and John Dacquisto. "Maglevs: The Future of Flying Trains – USC Viterbi School of Engineering." *Illumin Magazine*, 6 Dec. 2011, illumin.usc.edu/maglevs-the-future-of-flying-trains/.

Cruzan, Jeff. "Euler's Method." *Xaktly.com*, 2024, xaktly.com/EulersMethod.html.

Dawkins, Paul. "Differential Equations - Euler's Method." *Tutorial.math.lamar.edu*, 3 Dec. 2018, tutorial.math.lamar.edu/Classes/DE/EulersMethod.aspx.

Fiveable. "Superconductors in Transport | Superconducting Devices Class Notes." *Fiveable.me*, 2025, library.fiveable.me/superconducting-devices/unit-9.

Guerrieri, Marco. "High-Speed Railways, Maglev and Hyperloop Systems." *Springer Tracts in Civil Engineering*, 1 Jan. 2023, pp. 217–227, https://doi.org/10.1007/978-3-031-24030-0_12.

Hamed Yazdanpanah, and Mohammad Hosseini. "A Novel Method on Ion Propulsion Using Optimized Fuel Consumption." *4th International Conference on Science and Engineering*, 3 May 2016, www.researchgate.net/publication/354339074_A_Novel_Method_on_Ion_Propulsion_Using_Optimized_Fuel_Consumption?_tp=eyJjb250ZXh0Ijp7ImZpcnN0UGFnZSI6Il9kaXJlY3QiLCJwYWdlIjoiX2RpcmVjdCJ9fQ.

Herman, Edwin. "15.5: Divergence and Curl." *Mathematics LibreTexts*, 12 May 2018, math.libretexts.org/Courses/Monroe_Community_College/MTH_212_Calculus_III/Chapter_15%3A_Vector_Fields_Line_Integrals_and_Vector_Theorems/15.5%3A_Divergence_and_Curl.

Japan Rail Pass. “The Japanese Maglev: World’s Fastest Bullet...” *Japan Rail Pass Blog*, Japan Rail Pass - JRailPass.com, 27 Feb. 2017, www.jrailpass.com/blog/maglev-bullet-train.

JRCentral. “About the Vehicle | SCMAGLEV | Central Japan Railway Company.” *Scmaglev.jr-Central-Global.com*, scmaglev.jr-central-global.com/about/design/.

Kogar, Anshul. “Meissner Effect as Amplified Atomic Diamagnetism.” *This Condensed Life*, 4 June 2020, thiscondensedlife.wordpress.com/2020/06/04/meissner-effect-as-amplified-atomic-diamagnetism/.

K&J Magnetics. “Halbach Arrays | K&J Magnetics Blog.” *Kjmagnetics.com*, 2025, www.kjmagnetics.com/blog/halbach-arrays?srsltid=AfmBOooRw-EQP04IU7VJYNuGJDRIn6ISZLzx7IQDaDX8vsOdUZOsRr27. Accessed 20 Aug. 2025.

Magnetic Innovations. “DC Motor, How It Works | DC Adventages | Magnetic Innovations.” *Magnetic Innovations*, 2023, www.magneticinnovations.com/faq/dc-motor-how-it-works/.

Neumann, Erik. “MyPhysicsLab Runge-Kutta Algorithm.” *Www.myphysicslab.com*, www.myphysicslab.com/explain/runge-kutta-en.html.

Nykamp, Duane. “The Idea of the Curl of a Vector Field - Math Insight.” *Mathinsight.org*, mathinsight.org/curl_idea.

Powell, J. P., and R. Palacín. “Passenger Stability within Moving Railway Vehicles: Limits on Maximum Longitudinal Acceleration.” *Urban Rail Transit*, vol. 1, no. 2, June 2015, pp. 95–103, rd.springer.com/article/10.1007/s40864-015-0012-y, https://doi.org/10.1007/s40864-015-0012-y.

Singh, Timon. “Japan’s New 310 MPH “Floating” Maglev Bullet Train Hits the Tracks for Testing.” *Inhabitat - Green Design, Innovation, Architecture, Green Building | Green Design & Innovation for a Better World*, 8 June 2013, inhabitat.com/japans-new-310-mph-floating-maglev-bullet-train-hits-the-tracks-for-testing/. Accessed 20 Aug. 2025.

Sonier, J.E. “2.1.1 the London Penetration Depth.” *Triumf.ca*, 2025, cmms.triumf.ca/theses/Sonier/PhD/node7.html.

Stanford. “List of Superconductors and How They Work.” *Samaterials.com*, 2025, www.samaterials.com/content/list-of-superconductors-and-how-they-work.html.

Stanford Magnets. “Halbach Arrays: The Ingenious Magnetic Arrangement You Need to Know about | Stanford Magnets.” *Www.stanfordmagnets.com*, 16 Apr. 2024, www.stanfordmagnets.com/halbach-arrays-the-ingenious-magnetic-arrangement-you-need-to-know-about.html.

VisCircuit. “Euler vs Runge-Kutta Methods.” *Medium*, 31 May 2024, medium.com/@viscircuit/euler-vs-runge-kutta-methods-de34565bf0cf. Accessed 20 Aug. 2025.

Walpole, Sarah Catherine, et al. “The Weight of Nations: An Estimation of Adult Human Biomass.” *BMC Public Health*, vol. 12, no. 1, 18 June 2012, bmcpublichealth.biomedcentral.com/articles/10.1186/1471-2458-12-439, https://doi.org/10.1186/1471-2458-12-439.

Zoke Automation. “Linear Induction Motor(LIM) vs Linear Synchronous Motor(LSM).” *Zoke Automation*, 24 Feb. 2025, zoke-automation.com/blog/linear-motor/linear-induction-motor-vs-linear-synchronous-motor/. Accessed 12 Feb. 2026.